

Status: Conceptual Document — Application Specification

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Relationship to 8PF: Application specification consistent with 8PF v9.1 Lite. All terminology and notation aligned with 8PF conventions.

1. Executive Summary

This document defines an architecture for the long-term preservation of digital objects, prioritizing Lineage over Encapsulation. By separating the bit-level data stream (the "Baton") from its historical and interpretative record (the "Race"), the architecture achieves a survivability profile that resists bit-rot, format obsolescence, and malicious destruction across centuries.

The Baton carries the content. The Race carries the meaning. The binding between them is not an internal format structure but the documented act of transmission: the lineage event.

2. Core Components

2.1 The Baton — Raw Pixel Stream

The Baton is the unstructured data content of the preserved object. It is a headless, uncompressed stream of pixel values.

Properties:

Property Specification

Format Raw, no header, no magic numbers, no internal metadata

Structure Sequential pixel values, row by row, channel by channel

Bit depth 8-bit or 16-bit intensity values

Damage behaviour Strictly local. Bit-rot affects only the corrupted pixels. All other pixels remain perfectly readable

Failure mode None in the data body. No index to corrupt, no compression to cascade errors, no cross-reference table to break

Unlike standard preservation formats (PDF/A, TIFF, JPEG 2000), the Baton has no internal structure that can fail catastrophically. It accepts that damage will occur and ensures that surviving data remains fully usable.

2.2 The Race – The Lineage Record

The Race is the chronological record of the object's custody and identity. It is composed of a sequence of lineage states (t_0 , t_1 , t_2 ... t_n) that together establish the object's authenticity, provenance, and interpretability.

Each state record carries:

- The complete t_0 record (the foundational declaration)
- The current state of the object at that point in the lineage
- Lineage metadata (chain position, parent reference, transmission data)
- Integrity proof (checksum binding all components)

The Race is governed by the four lineage predicates of 8PF: Origin (O), Inheritance (I), Descent (D), and Sanctioned (S). Continuity of the Race (L) holds only while all four predicates are satisfied across every chain.

3. The Foundational Primer — tf0

Every preserved object begins with a tf0 (Trusted Foundational Zero) record. This is the "Primer" that teaches any future finder how to interpret the raw bits. It is the minimal mathematical key required to reconstruct a viewable image from the Baton.

3.1 tf0 Record Schema

Field Type Description

declarant Identity Named human with standing who declared foundational inception

timestamp DateTime When the declaration was made

image_width Numeric Pixel columns

image_height Numeric Pixel rows

bit_depth Numeric Bits per channel (8 or 16)

channel_count Numeric Number of colour channels

colour_space String sRGB, Adobe RGB, linear, etc.

byte_order String Little-endian or big-endian

channel_order String RGB, BGR, RGBA, etc.

baton_inception_hash Hash Cryptographic hash of the raw Baton at the moment of tf0 declaration. This is the binding field that ties the Baton to the Race.

content_description String What the image depicts, its significance, its provenance narrative

geographic_origin String (Optional) Descriptive region or coarse location of object origin. May be omitted or expressed at low granularity where precision creates risk to communities, individuals, or sensitive materials

witness_set Array[Identity] (Optional) Other humans attesting to the declaration

declarant_signature Cryptographic Binding of declarant to content

tf0_checksum Hash Hash of all above fields. Used for integrity verification of the tf0 record itself.

Note on geographic_origin: This field is optional. Where precise coordinates pose a risk – indigenous cultural records, politically sensitive materials, personal archives of at-risk communities – the field should either be omitted or expressed as a descriptive region at coarse granularity (e.g., "Pacific Northwest" rather than specific coordinates). The archivist must assess the safety of the declaring community when populating this field.

3.2 tf0 Properties

Per 8PF §0.2:

- Object-primary at declaration: tf0 is evaluated as an object state, not as a point in a chain. It is the singular exception to lineage-primary governance.
- Biologically grounded: Requires a human being with standing. No technical system can substitute.
- Drift tolerance $B = 0$: Foundational inception is exact by definition. All subsequent drift is measured against tf0.

4. The Propagation Axiom

In accordance with the LOCKSS principle (Lots of Copies Keep Stuff Safe) and 8PF §8.1, the tf0 record MUST be propagated immutably to every subsequent state in the lineage.

Rule: Every t0, t1, t2... tn carries the complete and verifiable tf0 record. No handoff is valid without propagation of the foundational primer. Mismatch between any two state records constitutes ¬L via Origin (O) predicate failure.

4.1 Verification

At any state in the lineage, integrity is verified by:

1. Internal: Recompute tf0_checksum from stored fields and compare
2. Cross: Compare tf0_checksum against any other state in the same lineage
3. Chain: Traverse lineage and confirm all state records carry identical tf0 records

4.2 Distributed Witnessing

With propagation, every subsequent state is a witness to foundational inception. The lineage itself attests to its own origin. Loss of all state records is lineage extinction; recovery is possible from any single surviving record. A single surviving Primer can teach a future finder how to interpret any surviving Baton fragments from the same lineage.

5. The Binding: Lineage Event

The Baton (raw data) and the Race (lineage record) are not bound by internal format structure. They are bound by the documented act of transmission: the lineage event.

5.1 Binding Mechanism

At each transmission event ($t_n \rightarrow t_{n+1}$):

1. The Baton is transferred directly from custodian to custodian
2. The tf_0 record is propagated immutably to the new state
3. The current state is appended to the new state record, including:
 - Hash of the Baton at time of handoff
 - Storage location or medium identifier
 - Identity of the receiving custodian
 - Timestamp of handoff
4. All four lineage predicates are evaluated and recorded:
 - Origin (O): tf_0 declaration acknowledged
 - Inheritance (I): Direct transfer verified; no silent substitution
 - Descent (D): Receiving custodian accepts as same lineage
 - Sanctioned (S): Receiving custodian holds legitimate authority

5.2 Integrity Verification

At any state in the lineage, the binding between Baton and Race is verified by:

1. Recompute hash of the stored Baton
2. Compare against the $baton_inception_hash$ recorded in the tf_0 record (for inception verification) and against the current state hash (for ongoing integrity)
3. Verify that the tf_0 record matches all other state records in the lineage

Mismatch constitutes -L, as specified in 8PF §8.5.

6. Storage Architecture

6.1 "Separated but Linked" Model

The Baton and the Race are stored on logically separate systems to prevent single-point failures. They are bound by the baton_inception_hash stored within the tf0 record and the current Baton hash stored within each state record.

Storage tiers:

Tier	Baton Storage	Race Storage	Applicability
Ideal	Separate physical media, separate geographic locations	Separate physical media, separate geographic locations	National archives, well-resourced institutions
Acceptable	Separate storage systems, same location	Separate storage systems, same location	Mid-sized archives, university collections
Minimum	Logically separate storage, same system	Logically separate storage, same system	Small archives, community groups, individual creators

All tiers maintain the principle of logical separation. The binding via the Baton hash ensures that the relationship between Baton and Race is verifiable regardless of storage configuration.

6.2 Rationale

Separation ensures that a single storage failure cannot destroy both the content and the means to interpret it. If the Baton degrades, the Race survives to document what was lost and how to interpret what remains. If Race records are lost but a single state record survives anywhere, the full tf0 record and lineage history are recoverable. If only the Baton survives without any state record, the raw pixels remain but are uninterpretable — this is the failure mode the architecture is designed to prevent.

7. Threat Model

This architecture treats transparency as the primary defense. By avoiding encryption and proprietary formats, it ensures the Primer is never locked behind inaccessible keys.

Threat Mitigation Strategy

Bit-Rot Local error isolation via headless raw format. Damage affects only corrupted pixels. No cascading failure.

Format Obsolescence Mathematical Primers (tf0). No proprietary decoder required. Dimensions, bit depth, and colour space are stated in human-readable fields.

Malice / Vandalism Hyper-redundancy via propagated state records. Tampering with one record is detectable by cross-verification against all others.

Loss of Custody Geographic referencing in state records. Descriptive origin and custody trail enable future finders to reconstruct provenance.

8. Degradation Scenarios

8.1 Baton partially degraded

- Lineage records intact. Hash comparison detects degradation.
- Surviving pixel data remains viewable using tf0 Primer.
- Metadata remains fully available from any state record.
- L holds if degradation is documented and predicates remain satisfied.

8.2 Baton completely lost

- Lineage records preserve the tf0 declaration and all prior handoff records.
- Image data is irrecoverable.
- The record of what existed, who declared it, and what it meant survives.
- A new lineage may be declared by a future finder, with the lost object as an ancestor, not a predecessor (see Baton and the Race §7).

8.3 All state records lost, Baton survives

- Raw pixels exist but are uninterpretable without external knowledge.
- Dimensions, colour space, and meaning are unknown.
- This is the failure mode the architecture is designed to prevent through propagation.

8.4 Single state record survives, Baton lost

- The metadata endures in the surviving record.
- No image data remains viewable.
- The tf0 declaration provides sufficient information for a future finder to understand what was preserved and why.
- Restart is possible in principle, even if the original image cannot be reconstructed.

9. Relationship to 8PF

This architecture is a direct application of:

- 8PF §0.1: tf0 as the singular, human-declared starting point
- 8PF §8.1: The Propagation Axiom, requiring immutable tf0 record replication to every subsequent state
- 8PF §8.4: Cross-verification of tf0 records across all states in the lineage
- 8PF §8.6: Distributed witnessing, where every state attests to foundational inception
- Baton and the Race §7: The restart caveat, where a surviving record enables a future finder to declare a new lineage informed by the preserved one

No new axioms are introduced. No existing axioms are modified. This specification is an application pattern, not a framework extension.

10. One-Sentence Summary

Store the image as raw pixels so that damage stays local; store the meaning as a tf0 record propagated to every handoff so that interpretation survives as long as the lineage does; bind them not by format but by the documented act of transmission — so that even one surviving fragment tells the story well enough to begin again.